SelectNorm: Technical Dossier (not submitted)

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Working and verification dossier for "SelectNorm: A Disagreement-Factorized Analysis of Per-Input Normalization for Arabizi with Frozen Encoders" (ACIIDS 2027). Values are copied unchanged from the archived tables; table and figure numbers in the main paper refer to that paper.

Table S1. Paired differences in macro-F1 (SelectNorm-adapt minus baseline, $\lambda = 0$, seed-averaged) with 95% paired-bootstrap intervals (10,000 resamples of test sentences). *: Holm-corrected $p < 0.05$ over the protocol's 20 comparisons per encoder. †: the interval lies inside $(-1, +1)$, the protocol's relevance margin; reading this as equivalence is post hoc and not corrected for multiplicity. Intervals without either mark are inconclusive about both difference and equivalence.

Enc.	Set	vs. B1	vs. B3	vs. B4	vs. SN-src
mE5	S_3	-0.71 [-1.25, -0.21]	-0.67 [-1.21, -0.17]	-0.07 [-0.27, +0.13]†	-0.03 [-0.11, +0.06]†
	S_4	-0.31 [-0.80, +0.17]†	-0.23 [-0.72, +0.25]†	-0.03 [-0.25, +0.18]†	-0.04 [-0.16, +0.08]†
	S_5	+0.39 [-0.22, +1.01]	+0.39 [-0.22, +1.01]	-0.14 [-0.57, +0.28]†	-0.17 [-0.47, +0.11]†
	S_6	+0.76 [+0.23, +1.30]	+0.75 [+0.23, +1.29]	+0.02 [-0.22, +0.27]†	-0.04 [-0.18, +0.11]†
XLM-R	S_3	+0.99 [+0.01, +2.02]	+0.91 [-0.07, +1.91]	+0.28 [+0.04, +0.52]†	+0.07 [-0.02, +0.17]†
	S_4	-0.66 [-1.20, -0.11]	-0.60 [-1.14, -0.04]	-0.04 [-0.18, +0.10]†	+0.02 [-0.04, +0.09]†
	S_5	-0.01 [-0.42, +0.45]†	+0.09 [-0.34, +0.56]†	-0.18 [-0.41, +0.03]†	-0.25 [-0.45, -0.07]*†
	S_6	-3.76 [-4.58, -2.93]*	-3.77 [-4.58, -2.94]*	-0.23 [-0.42, -0.03]†	+0.53 [+0.33, +0.73]*†
LaBSE	S_3	-0.19 [-0.75, +0.35]†	-0.24 [-0.73, +0.23]†	+0.03 [-0.19, +0.24]†	+0.03 [-0.08, +0.14]†
	S_4	-0.37 [-1.01, +0.28]	-0.35 [-0.95, +0.25]†	+0.15 [-0.06, +0.34]†	+0.01 [-0.11, +0.14]†
	S_5	+0.62 [-1.20, +2.44]	+0.57 [-1.17, +2.32]	-0.07 [-0.60, +0.47]†	+0.34 [-0.01, +0.71]†
	S_6	+5.46 [+4.44, +6.50]*	+5.04 [+4.05, +6.04]*	+0.32 [+0.06, +0.58]†	+0.54 [+0.30, +0.79]*†
MARBERTv2	S_3	-0.01 [-0.65, +0.64]†	-0.16 [-0.77, +0.48]†	+0.03 [-0.18, +0.23]†	-0.09 [-0.20, +0.01]†
	S_4	+0.57 [+0.08, +1.06]	+0.16 [-0.37, +0.68]†	+0.06 [-0.10, +0.22]†	+0.00 [-0.07, +0.08]†
	S_5	+0.58 [-0.45, +1.64]	+0.58 [-0.45, +1.64]	-0.11 [-0.63, +0.40]†	-0.01 [-0.36, +0.35]†
	S_6	-0.53 [-1.15, +0.08]	-0.61 [-1.22, -0.01]	+0.12 [-0.19, +0.43]†	-0.01 [-0.15, +0.12]†

Table S2. As Table 2 of the paper, with f also trained on synthetic Latin renderings (+SYN, exploratory regime).

	mE5+syn				**XLM-R+syn**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	80.9$^{\triangledown}$	65.6$^{\triangledown}$	69.1	**61.5**	69.0$^{\triangledown}$	59.8$^{\triangledown}$	64.7	59.7
B1 always normalise	81.6	67.5	72.6	58.0$^{\triangledown}$	72.8	63.2	62.9	59.1
B2 random, matched rate	81.2	66.5	70.4	61.0	70.8	61.6	63.9	59.7
B3 confidence threshold	83.0	68.7	**73.2**	60.7	74.5	63.0	64.7	60.3
B4 direct gain	**83.8**	**68.9**	72.4	60.9	**74.5**	63.8	65.1	60.6
B4b per-view loss	83.8	68.9	72.5	60.8	74.3	63.8	65.0	60.5
SelectNorm-src	83.6	68.8	72.9	61.1	74.3	63.5	**65.5**	60.4$^{\triangledown}$
SelectNorm-adapt	83.7	68.9	73.0	61.1	74.3	63.5	65.4	60.6
Bayes threshold $\lambda\bar{c}$	83.7	68.8	73.2	61.1	74.5	**63.9**	65.0	**60.8**
R two-view ensemble†	83.7	68.9	74.7	60.5	75.8	64.0	65.6	60.7
B6 oracle‡	91.7	80.9	85.7	71.4	86.3	79.0	80.6	73.5
	LaBSE+syn				**MARBERTv2+syn**			
	S_3	S_4	S_5	S_6	S_3	S_4	S_5	S_6
B0 original	76.6$^{\triangledown}$	64.9$^{\triangledown}$	71.3	**62.4**$^{\blacktriangle}$	77.0$^{\triangledown}$	64.2$^{\triangledown}$	57.0	50.1$^{\triangledown}$
B1 always normalise	83.7	67.8	74.3	58.7$^{\triangledown}$	81.8	67.8$^{\triangledown}$	59.3	**60.7**$^{\blacktriangle}$
B2 random, matched rate	80.3	66.3	72.9	59.9	80.8	67.3	59.1	58.3
B3 confidence threshold	**84.3**	68.2	74.3	60.4	82.9	68.7	59.3	59.5$^{\blacktriangle}$
B4 direct gain	84.3	**68.5**	74.4	59.7	82.5	68.7	59.6	59.3$^{\blacktriangle}$
B4b per-view loss	84.1	68.5	**74.8**	59.6	82.3	68.7	59.6	58.7
SelectNorm-src	84.1	68.3	74.7	60.0	82.5	68.8	59.8	59.0
SelectNorm-adapt	84.2	68.2	74.6	60.0	82.6	68.9	**59.9**	58.9
Bayes threshold $\lambda\bar{c}$	84.2	68.2	74.8	60.1	**83.0**	**69.2**	59.8	58.6
R two-view ensemble†	83.5	68.8	76.6	61.3	83.6	69.0	59.3	57.3
B6 oracle‡	92.9	81.1	86.8	73.1	92.6	81.6	72.3	69.3

Table S3. Routing behaviour (mE5, $\lambda = 0$): selection rate, harmful transformations per 100 inputs, share of oracle opportunities captured, and estimated cost relative to B0. Help/harm: number of inputs where normalisation corrects/introduces an error.

	S_3 (help 376, harm 114)				S_4 (help 538, harm 402)				S_5 (help 124, harm 89)				S_6 (help 407, harm 456)			
System	Sel. %	Harm	Capt. %	Cost	Sel. %	Harm	Capt. %	Cost	Sel. %	Harm	Capt. %	Cost	Sel. %	Harm	Capt. %	Cost
B1	100.0	8.0	100.0	0.98	100.0	14.0	100.0	0.97	100.0	12.2	100.0	0.98	100.0	10.9	100.0	1.04
B3	98.9	8.0	99.7	1.97	99.1	14.0	99.5	1.97	99.5	12.2	100.0	1.97	98.5	10.9	99.9	2.03
B4	90.2	7.6	96.1	1.89	88.8	13.1	93.8	1.87	86.4	10.9	95.5	1.84	66.8	7.9	76.9	1.70
SN-adapt	89.0	7.6	95.7	1.88	87.0	13.0	93.3	1.86	85.7	11.3	96.8	1.83	65.9	8.3	80.9	1.69
Oracle	26.4	0.0	100.0	1.26	18.7	0.0	100.0	1.18	17.0	0.0	100.0	1.17	9.7	0.0	100.0	1.10

Table S4. 95% intervals for the mechanism measurements of Table 3 of the paper (1,000 bootstrap resamples of test sentences, seed-averaged within each resample; estimators not refitted).

Enc.	Set	AUC_π	AUC_ρ	$\mathrm{AUC}_\rho^{\mathrm{tgt}}$	gap (pts)	Δ_{src}	Δ_{tgt}
mE5	S_3	[0.65, 0.71]	[0.60, 0.72]	[0.58, 0.69]	[-0.8, +6.8]	[-1.0, -0.2]	[-0.8, +0.7]
	S_4	[0.62, 0.66]	[0.52, 0.60]	[0.54, 0.61]	[+15.2, +21.9]	[-0.8, +0.1]	[-1.2, +1.0]
	S_5	[0.57, 0.66]	[0.55, 0.71]	[0.49, 0.64]	[+7.7, +21.1]	[-0.2, +0.5]	[-1.5, +2.9]
	S_6	[0.73, 0.76]	[0.50, 0.57]	[0.52, 0.60]	[+16.7, +23.5]	[-1.7, +0.6]	[-0.7, +0.9]
XLM-R	S_3	[0.61, 0.67]	[0.66, 0.76]	[0.63, 0.73]	[-0.8, +6.2]	[+0.1, +2.0]	[-0.3, +2.0]
	S_4	[0.64, 0.68]	[0.53, 0.60]	[0.53, 0.61]	[+8.0, +14.6]	[-0.9, +0.0]	[-0.2, +0.8]
	S_5	[0.65, 0.74]	[0.35, 0.53]	[0.50, 0.69]	[+4.5, +18.0]	[+0.0, +0.7]	[-2.1, +0.8]
	S_6	[0.71, 0.74]	[0.41, 0.48]	[0.63, 0.70]	[-10.9, -4.7]	[-4.0, -2.5]	[+0.6, +1.7]
LaBSE	S_3	[0.66, 0.72]	[0.55, 0.67]	[0.47, 0.60]	[-4.5, +3.7]	[-0.4, +0.6]	[-0.6, +0.3]
	S_4	[0.63, 0.67]	[0.54, 0.61]	[0.55, 0.62]	[+3.0, +8.9]	[-1.0, +0.2]	[-1.2, +0.3]
	S_5	[0.61, 0.69]	[0.52, 0.66]	[0.48, 0.62]	[-2.9, +9.6]	[-1.3, +2.1]	[-2.0, +2.4]
	S_6	[0.67, 0.70]	[0.54, 0.60]	[0.61, 0.68]	[+13.0, +18.5]	[-3.8, -1.3]	[+0.2, +1.3]
MARBERTv2	S_3	[0.66, 0.72]	[0.59, 0.69]	[0.55, 0.66]	[-2.2, +5.3]	[-0.4, +0.6]	[-1.2, +0.5]
	S_4	[0.69, 0.73]	[0.58, 0.65]	[0.61, 0.68]	[+10.0, +15.7]	[-0.1, +0.9]	[+1.6, +3.9]
	S_5	[0.70, 0.77]	[0.55, 0.69]	[0.47, 0.62]	[+7.0, +19.8]	[-1.9, +1.2]	[-1.3, +2.9]
	S_6	[0.69, 0.72]	[0.52, 0.58]	[0.59, 0.66]	[+1.7, +7.1]	[-0.9, +0.3]	[+0.3, +1.8]

Table S5. Sensitivity of the oracle reference (range over the four sets). "Views headroom": oracle over x and $T(x)$ minus the best view; other columns: the same for two classifiers differing only in their training sample (bootstrap as in Table 3 of the paper, 80%/50% subsamples, or bootstrap with $C \times 3$ or $C/3$), applied to x; mean of 10 pairs.

	views	reference headroom (two resampled classifiers)				
Enc.	headroom	bootstrap	80% sub.	50% sub.	boot., $3C$	boot., $C/3$
mE5	8.0–14.1	6.1–9.1	2.9–4.0	4.8–6.8	8.0–10.9	4.2–7.3
XLM-R	7.3–10.7	3.4–5.1	2.1–3.0	1.7–3.6	5.1–7.2	2.0–4.2
LaBSE	7.9–14.9	6.6–7.4	3.3–3.9	6.1–7.0	7.2–8.6	5.4–6.2
MARBERTv2	11.5–16.5	10.6–15.3	4.7–6.3	9.4–14.5	14.8–21.9	8.2–11.3

Table S6. Net effect of always normalising (help minus harm per 100 inputs) by input condition, mean over the four encoders, range in brackets. Conditions were fixed before inspection.

Condition (share of $S_3/S_4/S_5/S_6$)	S_3	S_4	S_5	S_6
French/English ≥30% (9/12/12/22)	+4.3 [+0,+11]	+4.3 [-2,+13]	+1.6 [-3,+5]	-0.1 [-9,+7]
not French/English ≥30% (91/88/88/78)	+17.0 [+15,+20]	+7.6 [+5,+9]	+6.5 [+5,+10]	+1.3 [-11,+12]
negation rule applied (8/12/26/6)	+14.0 [+6,+18]	+5.9 [-4,+20]	+4.0 [-4,+9]	+2.1 [-9,+15]
not negation rule applied (92/88/74/94)	+16.1 [+14,+18]	+7.4 [+6,+11]	+6.5 [+3,+11]	+1.0 [-11,+10]
lexicon coverage <50% (51/57/81/85)	+12.3 [+11,+14]	+4.5 [+2,+9]	+3.7 [+1,+6]	+0.8 [-11,+11]
not lexicon coverage <50% (49/43/19/15)	+19.7 [+17,+23]	+10.8 [+9,+14]	+15.1 [+8,+22]	+2.4 [-11,+13]
≤3 tokens (42/28/2/27)	+14.2 [+10,+16]	+8.4 [+0,+15]	+13.2 [-18,+35]	+0.7 [-10,+10]
not ≤3 tokens (58/72/98/73)	+17.2 [+13,+20]	+6.8 [+3,+10]	+5.7 [+4,+10]	+1.2 [-11,+11]

Table S7. Ablations (mE5, $\lambda = 0$, macro-F1, mean of five seeds). Rows with "always" are B1 with that normaliser.

Variant	S_3	S_4	S_5	S_6
SelectNorm-adapt (full)	81.3	66.9	69.8	53.7
– target re-estimation of $\hat{\pi}$	81.3	67.0	70.0	53.7
Bayes threshold instead of tuned	81.4	66.9	69.6	53.5
$2\hat{\rho} - 1$ only (no $\hat{\pi}$)	81.3	66.9	69.8	53.6
$\hat{\pi}$ only, tuned (label-free ranking)	81.9	67.2	69.5	53.1
confidence features only	82.0	67.2	69.4	53.0
– confidence features	81.5	66.9	69.6	53.6
– script features	81.6	67.1	69.8	53.4
– normaliser reliability	82.0	67.0	69.4	53.2
+ embedding PCA	82.0	67.2	70.9	53.8
gradient boosting	82.0	67.1	69.6	53.1
T_{lex} (no safeguards)	80.7	64.6	65.2	53.0
always T_{lex}	77.8	61.8	63.2	47.9
T_{rule}	76.7	61.8	61.4	53.3
always T_{rule}	75.1	58.8	58.8	47.7
T_{cov} (unknown tokens kept Latin)	75.0	65.7	70.8	54.1
always T_{cov}	75.0	65.8	70.7	54.0

Table S8. Normaliser accuracy against human Arabic-script transliterations: *development* results, not an independent test. TArC: 20% of sentences, not used for the lexicon $\mathcal{L}$ but used to design the safeguards. NArabizi: whole corpus, nothing fitted to it, but scored during development. "Foreign kept": precision/recall of keeping annotated French/English tokens in Latin script; negation: accuracy on circumfix-negated verbs. Audit: no hand-listed word occurs only in the development split; on the 87% of tokens no rule touches, word accuracy is 65.8%; with the splits' roles swapped, negation is kept in 94.4% and foreign precision/recall is 95/77.

	TArC development split (Tunisian)			NArabizi (Algerian, out of dialect)	
Strategy	Word acc.	Foreign kept (P/R)	Negation	Word acc.	Foreign kept (P/R)
T_{rule}	36.3	–/0	29.1	39.0	–/0
T_{lex}	69.5	–/0	54.4	47.6	–/0
T_{cons}	69.1	96/79	96.1	46.8	93/53

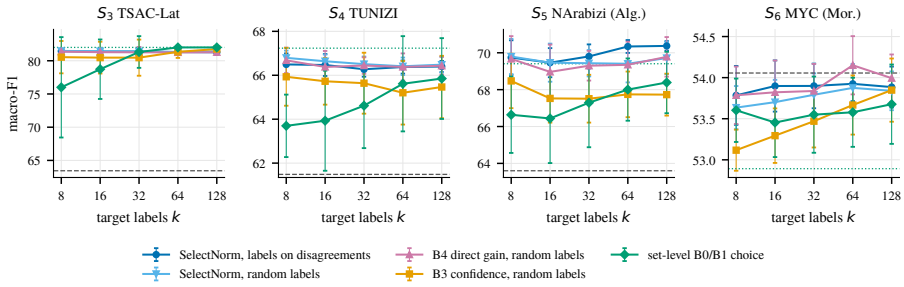


Fig. S1. Few-label adaptation (mE5): macro-F1 against the number k of labelled target sentences (5 draws × 5 seeds; error bars: s.d.). Dashed: B0; dotted: B1. "Set-level choice" picks B0 or B1 by accuracy on the k labels.

Implementation details

Encoders and software. Encoders are loaded at pinned revisions with SHA-256-verified weights: XLM-R base `e73636d` (MIT), mE5 base `d128750` (MIT), LaBSE `836121a` (Apache-2.0), MARBERTv2 `fe88db9` (no licence stated). Representations: mean of the last layer over non-padding tokens, at most 128 subwords, ℓ_2-normalised; mE5 inputs carry `query:`. Python 3, scikit-learn 1.9.1, NumPy 2.5.3; code, configuration, logs and the protocol are released.

Classifier f. ℓ_2-regularised logistic regression; $C \in \{0.01, 0.03, \ldots, 100\}$ selected by macro-F1 on TSAC-AR dev; temperature fitted on the same split by NLL. The +SYN regime adds the synthetic Latin rendering $G(x)$ of every training and dev sentence.

Router. $\hat{\pi}, \hat{\rho}$: logistic regression on standardised z ($C = 1$); 17 features: confidence, margin and entropy of p_0 (CONF); length, Latin and Arabic shares, digit-as-letter, emoji and upper-case rates (SCRIPT); lexicon, rule-fallback, ambiguity, protected, negation, laughter and changed-token rates and lexicon entropy of T (TREL). Policy split 80/20 of TSAC-LAT train with seed $1000 + s$; the 20% part tunes thresholds.

Few-label audit. `check_fewlabel.py` replays every targeted and random label draw of all eight encoder–regime runs (8,000 draws), saves the sampled source identifiers, and counts any sampled label from the scored fold, any shared source identifier or normalised text between sampled and scored sentences, and any targeted sentence that is not a disagreement: none. A rerun of `adapt.py` for mE5 with index logging matches the replay in all 1,000 draws and the archived results in all 2,500 rows. In 375 targeted draws (S_5, k=128) fewer than k disagreements were available.

Per-example exports. `results/export/per_example` gives, per encoder, regime and set, sentence and source identifiers, labels, both views' probabilities and every system's λ=0 prediction per seed. `scripts/reproduce_stats.py` (NumPy and pandas only) recomputes from them every difference, interval and p-value of Table 2 of the paper and Table S1 for all four encoders exactly.

Costs. c_T is the measured normaliser time (about 0.4% of a pass). c_0, c_1 are *estimated* per sentence by a per-encoder model $a + b \cdot$tokens fitted to measured batch-1 latencies of 300 sentences per set and view (e.g. mE5 $a = 19.6$ ms, $b = 0.32$ ms/token, $R^2 = 0.68$), as used in Fig. 2 of the paper and Table S3. `measured_cost.py` runs the deployed pipeline on raw text at batch size 1 with nothing cached (pinned encoder, the trained head refitted deterministically, which reproduces the cached probabilities exactly, normaliser, features, each seed's router and λ=0 threshold, second pass only when selected). On all 9,220 test sentences, none of the 184,400 live routing decisions (4 encoders × 5 seeds) or final predictions differs from the archived ones (probabilities within $1.6 \cdot 10^{-4}$). Timed back to back on 200 sentences per set: B0 22–31 ms; relative costs within −0.02 to +0.05 of the estimates for B1 and R, +0.03 to +0.07 for SelectNorm (features are not in the token model). mE5 was re-timed after a concurrent job overlapped it (both runs archived).